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The effect of the realism degree of avatars in social virtual worlds: The perspective of self-presentation

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ABSTRACT

Avatars, serving as a medium for users' self-presentation, play a significant role in bridging the real and virtual worlds. Limited research has considered users' desire to present their "hoped-for possible selves" when examining the effect of avatar design. This study addresses this gap by conducting four empirical investigations of the impact of avatar appearance design, particularly realism, on users' continuing intention to use virtual products. It was found that low-realism avatars, as opposed to high-realism ones, increased users' continuing use intention. Wishful identification and immersive experience serially mediated the avatar realism effect. The moderating role of users' familiarity with other avatars was also examined, suggesting that when users were familiar with other avatars, the impact of avatar realism diminished. These findings provide significant contributions to the existing literature and offer actionable guidance for the design of avatars within social virtual environments.

1. Introduction

Advances in digital technologies, including artificial intelligence (AI) and virtual reality (VR), have enabled real-time social interaction in virtual worlds from remote physical locations [1]. The virtual worlds allow users to freely explore, communicate, and participate in diverse activities [2,3]. A critical challenge for these virtual platforms is integrating users' real-world identities into virtual environments, which is essential for fostering immersive engagement. Avatars play a central role in addressing this challenge. A recent survey revealed that over 60 % of Gen Z and Millennials in the U.S. expressed interest in creating avatars to represent themselves in the metaverse [4], reflecting a desire to extend their social circles to the virtual environments. Responding to this trend, more companies are encouraging consumer engagement through avatars [5]. Such engagement spans the entire journey—from users selecting avatars that reflect their identities to interacting with and immersing themselves in virtual worlds via these avatars [6]. Notably, the initial phase of avatar selection is important for virtual platforms, particularly in terms of the avatar's degree of realism, as it may significantly impact users' interactions and experiences within virtual settings [1,6,7]. Platforms (e.g., Roblox and Zepeto) offer users extensive customization options to support self-presentation as value-added services, with 20 % of Roblox users updating their avatars daily [8]. This suggests that highly realistic avatars are not always preferred by users. Yet limited research has explored how users select avatars across different realism designs, as well as its downstream outcomes in virtual interactions.

Prior research on avatar realism has predominantly argued in favor of high-realism designs over low-realism ones (e.g., [1,9,10]). For example, Kim et al. [1] found that avatars with high realism could enhance user–avatar relationship and intention to use. These studies positioned users as observers or ordinary players and treated avatars as objects (e.g., "tools" or "characters") in virtual environments [11]. From this perspective, a match with reality—achieved through high avatar realism—is often considered preferable [11]. While some studies have recognized the potential benefits of low-realism avatars, their perspectives remain limited in avatars as objects [12]. For instance, Etemad-Sajadi [12] found that low realistic or cartoon-like avatars may lead to greater user satisfaction in commercial contexts. However, such framing may overlook users' fundamental desire to engage in self-presentation through avatars in virtual interactions. The present study aims to compare the effects of low versus high avatar realism from a self-presentation perspective and to explore the potential boundary conditions.

From a self-presentation perspective, avatar realism design can be viewed as a vehicle for presenting one's desired identity in virtual

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environments [13]. In this light, compared with high-realism avatars, low-realism avatars, though less aligned with users' real-world appearance, may provide users greater freedom to present their "hoped-for possible selves" [14]. Such freedom facilitates wishful identification, an important dimension of avatar identification that reflects the mental process by which individuals wish to imitate and even become the avatar in the virtual world [15,16]. Conversely, a high degree of realism reflects a strategy to present now selves that are similar to the real world, enabling a current match between users and their avatars (i.e., similarity identification; [11]). While such similarity is preferred when avatars are treated as objects, it may not maximize immersive experience when users actively use avatars for self-presentation. Therefore, we propose that wishful identification and immersive experience act as key mechanisms underlying the effect of avatars with low realism. As users increasingly identify with their avatars on a wishful level, they may experience a temporary loss of self-awareness and sense of time, the key aspects of immersion [17]. The sense of immersive experience, in turn, may foster users' continuing intention to use the virtual products. However, this effect may be attenuated by users' familiarity with other avatars in the environment, as individuals may adapt their self-presentation strategies based on social context [18]. When they are surrounded by known avatars, the freedom to present "hoped-for possible selves" may feel constrained [19], thus weakening the self-presentational benefits of low avatar realism.

In sum, this research aims to investigate whether and how a low-realism (vs. a high-realism) avatar appearance influences users' continuing intention to use virtual products, adopting a self-presentation lens. Four studies (Study 1–4) were conducted to examine the proposed effect. The results of Studies 1 and 2 show that low-realism avatars, compared with high-realism ones, significantly increase users' continuing usage intentions. Studies 2 and 3 further demonstrate that wishful identification and immersive experience serially mediate this effect. The moderating role of familiarity with other avatars was tested in Studies 3 and 4, showing that greater familiarity attenuates the influence of avatar realism.

This research contributes to the literature in the following ways: (1) It advances avatar realism literature from the perspective of self-presentation. (2) It extends the understanding of wishful identification from the traditional dyadic context (i.e., identification between "self" and "others") to a monadic one (i.e., identification between "real self" and "virtual self"). (3) It enhances the understanding of the impact of avatar realism across different contexts by considering the interplay between the real and virtual worlds. These findings offer actionable insights for marketing managers, providing practical guidance on the design of avatars' appearance.

2. Literature review

2.1. Self-presentation theory in virtual worlds

Self-presentation theory explains individuals' attempts to construct and display a favorable image in the eyes of others during social interaction [18]. Self-presentation has been extended from offline daily life (e.g., working and dating) to the virtual environments. This theory can help us understand the effect of avatar realism for several reasons. First, creating or selecting avatars in the virtual worlds serves as a means for individuals to develop, explore, and validate their desired self-images [20]. Furthermore, individuals use avatar choices strategically to convey specific identities to others. According to Goffman [18], self-presentation is based on situations, with individuals consciously changing their self-presentation strategies to suit their audiences and achieve different goals. The self-presentation process can be facilitated in virtual worlds, which provide expansive possibilities for presenting diverse aspects of the self [20]. Specifically, there are two strategies frequently used by individuals to present themselves in virtual worlds: extending the similar self in the real world (e.g., [21]) and pursuing a different or altered self that is unattainable or not experienced in reality [22,23]. These strategies reflect individuals' deliberate presentation of self-concepts.

According to Markus and Nurius [14], self-concepts involve two aspects: "now selves," describing established self-schemas that are known to others, and "possible selves," representing future-oriented images of what individuals may hope to become or be afraid of becoming. In offline life, the presence of a body may prevent people from presenting identities that are incompatible with their visible physical characteristics [13]. In contrast, technologies in virtual worlds offer customizable options, such as dressing and modifying the virtual body [20], allowing users to control and change their avatars' appearance and try out new roles. Therefore, the presentation of "hoped-for possible selves" can be primed in social virtual worlds, which helps explain the effect of low avatar realism.

2.2. Avatar design and self-presentation in virtual worlds

Virtual worlds are three-dimensional interactive environments where users engage through their avatars [24]. These worlds can be categorized into gaming virtual worlds and social virtual worlds [25]. Gaming virtual worlds are often characterized by fantasy-based narratives and quest-driven behavior. In contrast, social virtual worlds (e.g., Second Life) offer more reality-based thematic settings and encourage free social or economic interaction [26]. Our research focuses on individuals' self-presentation in social virtual worlds, which are defined as 3D environments that facilitate real-time interaction and provide extensive opportunities for engagement and experience [26].

Avatars, as a critical medium of self-presentation in social virtual worlds, serve as a digital bridge between the physical and virtual realms [13]. The avatar design has garnered significant research interest, including the core identity elements (e.g., race and gender) and peripheral form elements (e.g., body, face, hair, and style of clothes; [7]). Previous studies have explored how the attractiveness, similarity to users, and level of abstraction of avatar design influence users' perception [7,11,27] and behaviors [28]. Messinger et al. [7] posited that the avatars for self-presentation generally maintain core characteristics—such as human-like body proportions and racial traits—similar to real-world representations, while facial and body features tend to show different ranges of variation. Our research focuses on the changeable design—the realism of appearance, also known as form realism [5]—because it explicitly conveys users' identities and images. In social virtual worlds, avatars have a basic 3D human-like body (e.g., head, arms, hair, features), and the degree of realism can be morphed by changes in texture resolution or facial proportions. Thus, we focus on the change in facial design between cartoon-like and photo-like appearance.

Literature reveals inconsistent findings regarding the impact of avatar realism [5]. Some studies suggest that more realistic avatars enhance users' sense of social presence, increasing their willingness to engage [29]. Conversely, other research indicates that less realistic, cartoonish avatars may improve consumer satisfaction with retailers [12]. This debate may stem from two factors. First, previous studies have primarily positioned users as observers and avatars as "tools" or "characters" [11], while neglecting the self-presentation perspective [12,30]. Second, some studies have focused on overall visual characteristics, such as human/machine body form or human/caricatured body proportion, without considering the specific facial realism of the avatar design [1,30]. For example, Kim et al. [1] considered avatars with human-like body proportion as high-realism, and considered avatars with smaller and caricatured body proportion (such as Lego-like designs) as low-realism. Our research acknowledges the varying effects of avatar realism across different contexts and perspectives (e.g., [1,11,12]). In our context, we focus on the self-presentation perspective, which may provide fresh insight into the effect of avatar realism.

3. Hypothesis development

3.1. The main effect of avatar realism on continuing use intention

Avatar realism exists on a spectrum, ranging from simple abstract geometric forms to highly realistic representations with advanced texturing and shading [31]. Specifically, the distinction between high and low realism is characterized by: (1) the level of realism in visual cues (2D vs. 3D models; [32]) and (2) the authenticity and accuracy of the avatar's appearance [33]. Miao et al. [5] also concluded that low realism involves 2D and cartoon design, while high realism is distinguished by 3D and photorealistic design. Similarly, we define cartoon-like facial designs as low realism and photo-like designs as high realism, primarily distinguished by variations in texture resolution and facial proportions.

Importantly, continuing use intention is pivotal for the enduring success of a service. While acquiring early adopters is a crucial first step, it is the continuing use that guarantees longevity and ultimate success of the services/products [34]. In virtual world contexts, continuation has been identified as a key consequence that is frequently investigated [2]. The realism design of avatars has been proven to influence individuals' attitudes and behavior. For example, previous studies have found that low-realistic avatar characters can improve players' satisfaction with retailers, product attitudes, and purchase intentions [12]. The users want the avatar to represent part of "me" with a tendency to diverge rather than converge [35].

Accordingly, we propose that low-realism avatars may be preferred by users for self-presentation in social virtual worlds. This preference can be understood through the lens of self-presentation theory, which highlights that avatars can provide a platform for users to express altered or idealized versions of themselves that may not be possible in their real lives. This capability comes from the fluid and infinite possibilities of virtual environments and the customizable nature of avatars [13,36]. In this context, a low-realism avatar may enable self-presenters to experiment with different roles more effectively than a high-realism one. In contrast, high realism, with its excessive real-world information, may go against the individual's desire to present a hoped-for possible self in a virtual world experience [37]. As a representation of users within the virtual world, an avatar's degree of realism may affect their intention to continue using the platform. Therefore, we propose the following hypothesis:

Hypothesis 1. *The appearance design of the avatar with a low (*vs. *high) realism degree will increase users' continuing use intention.*

3.2. The mechanism of wishful identification and immersive experience

The design of avatars mirrors the virtual identity of users, fostering a distinct sense of identification with their avatars [6]. Avatar identification is defined as the degree to which users identify with an avatar as themselves [38], which allows users to replace or extend their real-world identities. It encompasses three dimensions: similarity identification, wishful identification, and embodied presence [19]. In this study, wishful identification is most relevant, for several reasons. First, similarity identification typically describes a current match (i.e., match with the reality), while wishful identification, a form of "idealization," involves users relating to their avatars in a more imagined manner [19, 39]. Second, embodied presence arises from users' sense of control over the avatar's movements and actions. However, this is not the central focus of our research, as we are primarily interested in avatars' appearance design. Traditionally, studies view wishful identification in a dyadic context, focusing on the relationship between the media audience and fictional characters [40–42]. For example, Lim et al. [41] examined the role of viewers' wishful identification with live streamers. This research aims to shift the focus to a monadic context—concentrating on users' wishful identification with their possible virtual selves.

Some evidence in the literature has demonstrated the relationship between avatar realism and users' wishful identification. The effect of low-realism avatars reflects the "magic mirror" effect—users may have stronger identification for avatars that reflect an imagined self rather than a realistic one [43,44]. For example, Kim et al. [43] found that when the avatars in virtual worlds are decorated with diverse/expressive items, users have higher wishful identification for their avatars. A low-realism design of avatars in social virtual worlds provides users with a degree of flexibility in their possible self-presentation [13]. When evaluating their avatars in the virtual worlds, some users stated that "Things I would not do or wear in the real world I am free to do/wear in the virtual world" ([26], p.333). This suggests that when users are faced with avatars of higher realism, the freedom to project a "possible self" may be diminished. Therefore, we proposed that avatars with low realism are more likely to foster wishful identification, as they better support the user's desire to explore imagined aspects of the self in social virtual worlds.

Immersive experience is a critical component of user engagement within virtual environments, reflecting the user's level of absorption, engagement, and engrossment [45]. Avatar identification, which involves merging one's identity with a virtual character, can lead to a temporary loss of self-awareness and sense of time, the key aspects of immersion [17,46]. Studies have demonstrated that a stronger identification with the avatar can significantly enhance users' perceived hedonism and facilitate both immersion and flow experiences [47,48]. For example, Soutter and Hitchens [48] demonstrated that an increase in a player's identification with their video game character would be related to an increase in their ability to enter a flow state. Thus, this study also links users' wishful identification with avatars and immersive experience in a social virtual world setting.

Meanwhile, previous research has shown that users' immersive experience in the virtual world is a key variable for subsequent positive attitudes and behavior, including satisfaction, loyalty, repeat intention, and continuing use [37,49,50]. For example, Roh et al. [50] found that flow state, an optimal moment in an immersive experience, would increase users continuing use intention in metaverse. Therefore, we propose that the impact of avatar realism on continuing use intention may be serially mediated by wishful identification and immersive experience. Specifically, the following hypothesis was proposed:

Hypothesis 2. *The appearance design of the avatar with a low (*vs. *high) realism degree will enhance users' wishful identification and then improve their senses of immersive experience in social virtual worlds, thereby increasing continuing use intention.*

3.3. The moderating effect of familiarity with other avatars

Self-presentation efforts are shaped not only by the self but also by the social contexts [18], prompting us to explore the boundary conditions of the low-realism effect. Familiarity with other avatars refers to the extent to which users recognize and are familiar with other users' avatars within a virtual environment, serving as an important social cue in influencing individuals' behavior [51]. First, individuals adopt different self-presentation strategies based on their perceived social contexts [18]. In unfamiliar environments, users feel less constrained by social expectations, allowing them to present more idealized versions of themselves. Second, when users are immersed in a virtual world, they tend to focus more on the current experience and reduce their connection with the outside world [52]. Therefore, individuals in an unfamiliar environment may have more opportunities to develop wishful identification. In contrast, when users in the virtual worlds are reminded of the real world or their attention turns to their real-life identity, it may diminish the user's desire to represent themself with an avatar [19] and interrupt their immersive experience [53].

That is, when users are familiar with other avatars in the virtual world, it may trigger associations with their real-life social relationships,

thereby weakening the influence of avatar realism. In other words, different realism degrees of virtual avatars may have no impact on wishful identification when individuals are highly familiar with other avatars in the social virtual world. Therefore, we propose the following hypothesis:

Hypothesis 3. *Familiarity with other avatars in social virtual worlds moderates the process by which avatar realism degree (high* vs. *low) affects users' wishful identification. When users are familiar with other avatars, the realism degree (high* vs. *low) of avatars will not affect their wishful identification.*

4. Research methods

4.1. Overview of studies

We conducted four studies to test our hypotheses, as summarized in Table 1. Studies 1–3 used scenario-based experiments with virtual avatars created by computer. Scenario-based experiments have been widely applied because of the advantages of controlling for potential bias, as well as the convenience of collecting data from large samples [54]. Specifically, Study 1 tested the main effect of avatar realism with a female avatar in a virtual park scenario (H1). Study 2 tested the mediating effect of wishful identification and immersive experience with a male avatar in a virtual exhibition scenario (H1 and H2). Study 3 tested the moderated mediation model using a female avatar in a virtual shopping mall scenario (H2 and H3). To further validate the moderating effect of familiarity with other avatars in a real-world context, Study 4 used avatars generated by AI based on real photos uploaded by participants to examine users' preferences for avatar realism in different familiarity contexts (H3). Through varying both the avatars and scenarios, the external validity of our research could be enhanced.

4.2. Study 1

4.2.1. Design

This study followed a single-factor (avatar realism: low vs. high) between-subjects experiment that aimed to examine the impact of avatar realism on continuing intention to use a virtual world product. One hundred subjects (M_{age} = 35.01; 51 % female) were recruited from the Wenjuanxing platform (www.wjx.cn), a well-established survey website in China boasting over 6.2 million registered members [55]. Through their professional sampling service, a random sampling method was used to select participants. Upon completion of the experiment, each participant received a financial incentive of 0.5 Chinese yuan. Participants were first introduced to the virtual worlds and avatars through an informational page and were then asked to imagine themselves experiencing a virtual park via an avatar. To be more practical, we also presented participants with multi-angle pictures of this virtual park (see Appendix A). Next, participants were randomly assigned to a high- or low-realism group and informed that "The following is your avatar in the virtual world," while being shown a female virtual avatar. According to the proposed characteristics of realism, we designed the low-realism avatar as a cartoon-like character, while the high-realism avatar was a photo-like one (see Appendix A).

Table 1
Overview of studies.

Study	Design	Scenario	Method	Results
Study 1	A single-factor (avatar realism: low vs. high) between-subjects design through an online scenario-based experiment	A female avatar in a virtual park	One-way ANOVA	Confirm H1
Study 2	Same as Study 1	A male avatar in a virtual exhibition	One-way ANOVA and Model 6 in PROCESS	Confirm H1 and H2
Study 3	A 2 (avatar realism: low vs. high) × 2 (familiarity with other avatars: familiar vs. unfamiliar) between-subjects design through an online scenario-based experiment	A female avatar in a virtual shopping mall	Model 83 in PROCESS and CB-SEM method	Confirm H2 and H3
Study 4	A quasi-field experiment	Avatars based on participants' photos in a virtual exhibition	A χ^2 test	Confirm H3

After the above experience, participants' evaluation of the realism degree of the avatar (Cronbach's α = 0.819; [56,57]) and continuing intention to use the virtual world product (Cronbach's α = 0.744; [43]) were measured (see Table 3). All measurements were on seven-point scales (1 = strongly disagree, 7 = strongly agree). Finally, participants were asked to answer various basic descriptive statistical questions.

4.2.2. Results

Manipulation test. The results of one-way ANOVA showed that the degree of realism was significantly different among different groups ($M_{\text{low degree of realism}} = 3.8$, $M_{\text{high degree of realism}} = 4.73$, $F(1,98) = 7.699$, $p < 0.01$). This indicates that participants in the high-realism group did perceive the avatars as having a higher degree of realism in design, confirming the validity of the independent variable manipulation.

Continuing use intention. A one-way ANOVA revealed a significant main effect of the avatar realism on continuing use intention. Participants in the low degree of avatar realism group showed higher continuing intention to use the virtual world product than those in the high degree of avatar realism group ($M_{\text{low degree of realism}} = 4.52$, $M_{\text{high degree of realism}} = 3.33$, $F(1,99) = 13.087$, $p < 0.001$). Thus, H1 was supported.

Consistent with H1, Study 1 proved that continuing use intention was stronger for users using an avatar with a lower degree of realism. To thoroughly investigate the underlying mechanism of the low avatar realism effect while repeating the results of Study 1, Study 2 further explored the serial mediating roles of wishful identification and immersive experience.

4.3. Study 2

4.3.1. Design

A single-factor (avatar realism: low vs. high) between-subjects experiment was conducted to test the serial mediating roles of wishful identification and immersive experience. With a data collection method similar to that of Study 1, 182 participants from the Wenjuanxing platform (M_{age} = 27.13; 48.35 % female) were recruited randomly. They were allocated to either the scenario with an avatar with low degree of realism or the one with an avatar with high degree of realism. The stimuli used for Study 2 were similar to those for Study 1. The virtual world scenario was changed to a virtual exhibition and the gender of virtual avatar was changed to male (see Appendix B). The realism manipulation of this male avatar was checked by a pre-test, as shown in Appendix C.

The experimental process also resembled Study 1's approach. The participants were asked to evaluate their continuing intention to use the virtual world product, wishful identification, and immersive experience, as well as the realism degree of the virtual avatar (see Table 3). The continuing use intention and avatar realism scales were the same as in Study 1. Wishful identification was measured using a 7-point item developed by Van Looy et al. [19] and Mancini et al. [[58]; Cronbach's α = 0.904]. Immersive experience was measured using a 7-point item

developed by Yim et al. [[45]; Cronbach's α = 0.896].

4.3.2. Results

Continuing use intention. One-way ANOVA findings showed a significant avatar realism effect. Participants' continuing intention to use the virtual world product was higher when the avatar realism was lower ($M_{\text{low degree of realism}} = 4.81$, $M_{\text{high degree of realism}} = 4.18$, F (1180) = 6.406, $p < 0.05$), which was consistent with the findings from Study 1. Thus, H1 was supported again.

Serial-mediation analysis. The serial mediating roles of wishful identification and immersive experience were tested using Model 6 in PROCESS (bootstrap sample: 5000; [59]). Avatar realism was identified as the independent variable, with wishful identification and immersive experience identified as serial mediators and continuing use intention as the dependent variable. Study 2 further indicated a significant avatar realism → wishful identification → immersive experience → continuing use intention serial-mediation process (β = −0.169, SE = 0.034, 95 % CI [–0.243, –0.106]). However, there was no direct effect between avatar realism and continuing use intention (β = –0.0297, SE = 0.0626, 95 % CI [–0.153, 0.939]). Therefore, wishful identification and immersive experience fully mediated the effect of avatar realism on continuing use intention, supporting H2 (Fig. 1).

According to the findings in Study 2, continuing use intention was significantly influenced by avatar realism (H1), matching the outcomes of Study 1. Furthermore, Study 2 verified the serial mediating effects of wishful identification and immersive experience (H2), which explained the underlying mechanism between the avatar realism and continuing use intention. Study 3 was conducted to further confirm H2 and investigate the moderating role of familiarity with others for the above mechanism.

4.4. Study 3

4.4.1. Design

A 2 (avatar realism: low vs. high) × 2 (familiarity with other avatars: familiar vs. unfamiliar) between-subjects design was adopted to test the moderating role of familiarity. With a data collection method similar to those in Studies 1 and 2, 310 subjects from the Wenjuanxing platform ($M_{\text{age}} = 34.4$ years; 46.8 % female) were recruited randomly and placed in one of the above four conditions. Table 2 shows the demographic information for each cell. Participants have no significant difference in gender, age, income, or education among the four groups.

A virtual shopping mall and a female virtual avatar, the same as in Study 1, were used in the stimuli. An episode task was used to prime participants' familiarity with other avatars, following a recent study [51]. We first introduced the corresponding scenario to participants, similarly to previous studies. Then participants in the familiar group were asked to recall the names of women they were familiar with in real life and to imagine encountering them in the virtual world. Meanwhile, participants in the unfamiliar group were asked to imagine encountering another virtual avatar. They were then asked to imagine themselves in a virtual shopping mall by watching some pictures of a virtual mall taken from different angles. All experimental groups received the same information descriptions and pictures (see Appendix D).

On the next page, participants were randomly assigned to two groups with different avatar realism degrees, as previous studies did. Finally, continuing use intention, immersive expeirence perception of the virtual world, wishful identification, and realism degree of virtual avatar were measured in the same way as in Studies 1 and 2 (see Table 3). Familiarity with others was measured with "Regarding other users in that virtual world, I think it's: 1 = not at all familiar / 7 = very familiar" [60].

4.4.2. Results

Manipulation test. Participants in the familiarity group confirmed that they were more familiar with others in the virtual world ($M_{\text{familiarity}} = 4.72$, $M_{\text{unfamiliarity}} = 2.78$, $F(1, 308) = 80.978$, $p < 0.001$). Thus, the manipulation of familiarity was confirmed.

Moderated mediation analysis. In this study, Hayes' moderated mediation model (Model 83, bootstrapping 5000 times; [59]) was used to test the moderating role of familiarity. The degree of avatar realism was used as an independent variable, the degree of familiarity (0 = unfamiliar, 1 = familiar) was used as a moderating variable, wishful identification and immersive experience were used as chain mediators, and continuing use intention was used as a dependent variable. The results showed that the moderated mediating effect of familiarity was significant ($b = 0.0186$, SE = 0.0109, 95 % CI: [0.0034, 0.0456]). Specifically, when the individual had a low degree of familiarity with others, the serial mediation effect of wishful identification and immersive experience was significant (indirect path effect = –0.0209, SE = 0.0101, 95 % CI: [–0.0449, –0.0055]). When the individual had a high degree of familiarity with others, the serial mediation effect of wishful identification and immersive experience was not significant (indirect path effect = –0.0023, SE = 0.0052, 95 % CI: [–0.0129, 0.0081]). The specific path coefficients are shown in Fig. 2.

Discussion. Study 3 reproduced the findings from Study 2, which confirmed the reliability of the experimental results, i.e., that the serial mediating effect of wishful identification and immersive experience was tested (H2), and the moderating role of familiarity with others was explored (H3). To enhance the robustness of our study, we further tested H2 and H3 in one model by the CB-SEM method with SmartPLS 4.0, because it is suitable for theory testing and confirmation. The results of the CB-SEM method are shown in Appendix E. Additionally, the first three studies all adopted virtual avatars generated by computers as stimuli. One of biggest advantages of using standard avatars in the experiments is to control the appearance and characteristics of the avatars precisely and minimize bias that may arise due to personal appearance differences, helping maintain consistency in experiments. However, in order to match the real-world usage scenario, we conducted a quasi-field experiment (Study 4) that allowed participants to upload personal photos and use AI technology to generate versions of avatars with different levels of realism.

4.5. Study 4

4.5.1. Design

This study aims to evaluate users' preference for degree of avatar

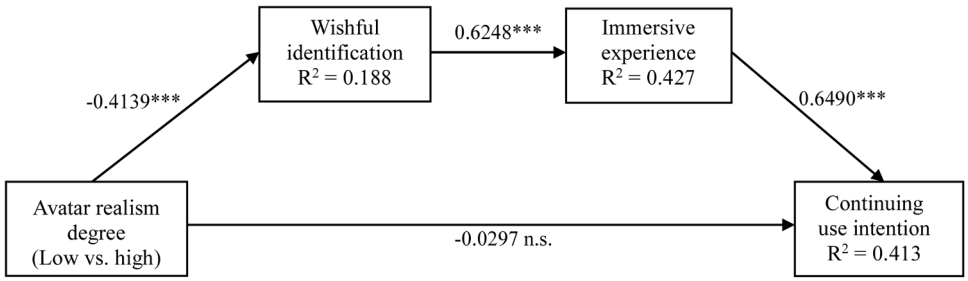


Fig. 1. Serial mediation effect in Study 2. *Notes:* ***$p < 0.001$; n.s.: non-significant.

Table 2
Demographics of participants ($N = 310$).

Category	Response	Percentage (%) of group				Percentage (%) of total	χ^2 test (p value)
		Group 1 (low realism, familiarity)	Group 2 (high realism, familiarity)	Group 3 (low realism, unfamiliarity)	Group 4 (high realism, unfamiliarity)		
Gender	Male	51.3	54.5	53.2	53.8	53.2	**0.980**
	Female	48.7	45.5	46.8	46.2	46.8	
Age	18–25	7.7	23.1	23.1	46.2	4.2	**0.298**
	26–35	23.0	24.8	24.2	28.0	51.9	
	36–45	29.4	25.0	25.7	19.9	43.9	
Income	<3000 CNY	26.9	19.5	24.7	26.9	24.5	**0.902**
	3000–5000 CNY	26.9	28.6	27.3	23.1	26.5	
	5000–10,000 CNY	24.4	23.4	29.9	28.2	26.5	
	10,000–20,000 CNY	10.3	13.0	10.4	14.1	11.9	
	>20,000 CNY	11.5	15.6	7.8	7.7	10.6	
Period of schooling	<9 years	16.7	14.3	10.4	21.8	15.8	**0.444**
	9–12 years	26.9	29.9	35.1	35.9	31.9	
	13–16 years	43.6	46.8	39.0	34.6	41.0	
	>16 years	12.8	9.1	15.6	7.7	11.3	

Table 3
Construct operationalization.

Constructs	Items	Reference(s)
Continuing use intention	• I intend to continue, not discontinue, using this virtual world • My intention is to continue using this virtual world, not others • If I could, I would like to continue using this virtual world	Kim et al. [43]
Immersive experience	• 1 = Not deeply engrossed; 7 = Deeply engrossed • 1 = Not absorbed; 7 = Absorbed • 1 = My attention was not focused; 7 = My attention was focused	Yim et al. [45]
Wishful identification	• If I could become like my avatar, I would • I would like to be more like my avatar • My avatar has characteristics that I would like to have	Van Looy et al. [19] Mancini et al. [58]
Degree of realism	• The avatar looks like a person • The avatar looks very realistic • The avatar looks very cartoonish (reverse)	Nowak & Rauh [57] Go & Sundar [56]
Familiarity with other avatars	• Regarding other users in that virtual world, I think it's: 1 = Not at all familiar; 7 = Very familiar	Mariconda & Lurati [60]

realism under familiar versus unfamiliar conditions through a quasi-field experiment. One hundred and two students from several universities in China, who were recruited via purposive and convenience sampling, participated in this experiment in exchange for 10 Chinese yuan. Prior to the experiment, participants were informed about the purpose of the study, their right to withdraw at any time, and the measures implemented to ensure confidentiality and anonymity.

Participants were initially requested to upload a personal photo without any stylistic restrictions. These photos were then submitted to the Jimeng AI (https://jimeng.jianying.com/), a tool developed by ByteDance in 2024 that enables users to generate images from uploaded photos and textual descriptions. Utilizing two distinct prompts, researchers generated avatars with low (vs. high) degree of realism (see Appendix F). Image matting techniques were employed to isolate the human figure, eliminating background interference. During the experiment, participants were briefed on the virtual worlds and avatars, followed by the completion of a questionnaire. In the questionnaire, all participants imagined that they were invited to attend a virtual exhibition through their own avatars, and they were randomly assigned to the familiar or unfamiliar condition. Under the familiar condition, participants were asked to list the names of three individuals they felt close to and imagine attending this virtual exhibition with them. In the unfamiliar condition, participants were informed that they would encounter strangers from around the world. Finally, participants were asked to select their preferred avatars and provide reasons in an open-ended question.

4.5.2. Results

Sample characteristics. Among the 102 participants, 5 failed to finish the experiment as required, leaving 97 valid samples (66 % female). All the participants were aged 18–30, as Gen Z and Millennials have most interest in creating avatars to represent themselves in the metaverse [4].

Preference for avatar realism within familiar versus *unfamiliar condition.* Out of a total of 97 valid participant samples, 48 were assigned to the familiar condition and 49 to the unfamiliar condition. Among the participants in the unfamiliar group, a majority of them (68.8 %) expressed a preference for avatars with a low degree of realism. In contrast, within the familiar group, the distribution was more evenly

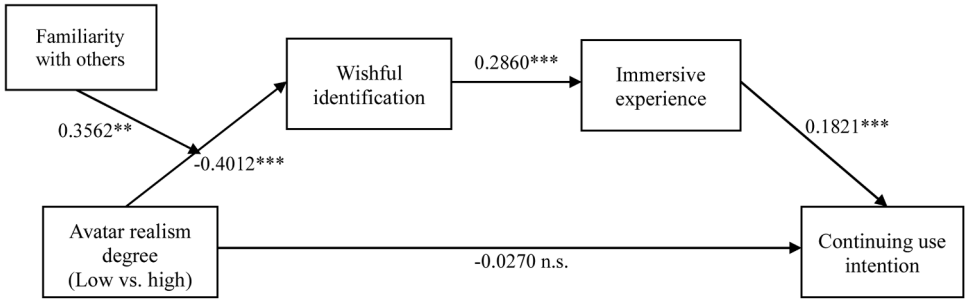


Fig. 2. Moderated meditation effect in Study 3. *Notes:* $^{**}p < 0.01$, $^{***}p < 0.001$; n.s.: non-significant.

split, with 46.9 % choosing avatars with low realism and 53.1 % choosing avatars with high realism (see Fig. 3). To statistically evaluate the difference in preferences between the two groups, a Pearson χ^2 test was conducted. The test revealed a significant difference in preferences for avatar realism between the unfamiliar and familiar groups (100 %, $\chi^2 = 4.727$, $p = 0.03$).

The findings of Study 4 indicated that participants in unfamiliar settings lean more toward avatars with a low degree of realism, while those in familiar settings show more balanced preferences. This result confirms the moderating role of familiarity from a real-world perspective, supporting H3 once again.

5. Discussion

Avatars, important self-presentation media connecting users and the virtual worlds, have vital impacts on individuals' experiences in social virtual worlds. Therefore, how to link users' real-world identities to virtual worlds through the avatar design has become an important issue for marketing practitioners and researchers. This research focuses specifically on the realism of avatar appearance, conducting four studies to examine the effect of avatar realism on users' continuing use intention and its mechanism.

5.1. Theoretical implications

This research makes several theoretical contributions. First, it enriches the literature on avatar realism from the perspective of self-presentation. Previous studies show inconsistent findings on the effect of avatar realism (e.g., [12,30]). Our results suggest that the lens of the user—as an observer or a self-presenter—plays a critical role in shaping the impact of avatar realism. Specifically, previous studies found that high realism might be preferred when users were observers [30]. However, avatars also serve as tools for self-presentation, allowing individuals to present their self-concepts in virtual worlds [13]. This research considers that users tend to pursue the hoped-for possible self (i.e., low realism) in virtual worlds, echoing Chakraborty et al.'s [[22], P. 4] claim that "there is an expansion of possibilities for consumers to discover new methods to express themselves."

Second, this research extends the understanding of wishful identification from the traditional dyadic context, which involves media audiences and fictional characters, to a monadic context within the realm of social virtual worlds. Building on Hefner et al.'s [40] call to explore monadic wishful identification, this study shifts the focus from the identification between "me" and "others" to that between users' real-world selves and their virtual selves. Specifically, this research finds that self-presenters can develop a greater wishful identification with low-realism (vs. high-realism) avatars. This can be attributed to the fact that unrealistic clues are more likely to help them construct a hoped-for possible self that aligns with their wishful identification in social virtual worlds.

Third, this research enhances the understanding of the impact of avatar realism across different contexts by considering the interplay between the real and virtual worlds. We specifically investigate the moderating role of familiarity with other avatars, revealing a critical dynamic: when users are familiar with other avatars, the effectiveness of avatar realism diminishes. This familiarity constrains users' motivation to "create a different identity," thereby reducing the appeal of low-realism designs. In essence, the potency of avatar realism is contingent upon the closeness (i.e., familiarity) or distance (i.e., unfamiliarity) of users' connections between the real world and the virtual world. Such exploration is important for comprehending self-presentation behaviors within virtual environments.

5.2. Managerial implications

This research offers actionable insights for businesses and designers seeking to optimize avatar design in social virtual worlds. First, the findings reveal that low-realism avatars can be more effective than high-realism ones in promoting users' continuing use intentions by fostering wishful identification and immersive experiences. Platforms might therefore consider prioritizing stylized, cartoon-like avatar options based on human-like features, especially for new users who seek self-presentation. For example, during onboarding, platforms could recommend low-realism avatars with messaging such as "Discover your ideal self," encouraging users to explore imaginative self-presentation.

Second, Studies 3 and 4 indicate that the influence of avatar realism diminishes when users are familiar with other avatars in the virtual environment. In such contexts, reminders of the real world may interrupt users' desire for self-presentation [19] and their immersive experience [53]. Therefore, a dynamic customization strategy—offering cartoon-like or photo-like avatars depending on users' social network strength—may be beneficial. For example, in unfamiliar environments (e.g., virtual exhibitions or open-world exploration), platforms might promote low-realism avatars. This approach can encourage users to explore their "hoped-for possible selves" [13], thereby enhancing wishful identification and immersion. In contrast, in environments involving familiar contacts (e.g., virtual workplaces or private social groups), avatar realism may have reduced the psychological impact.

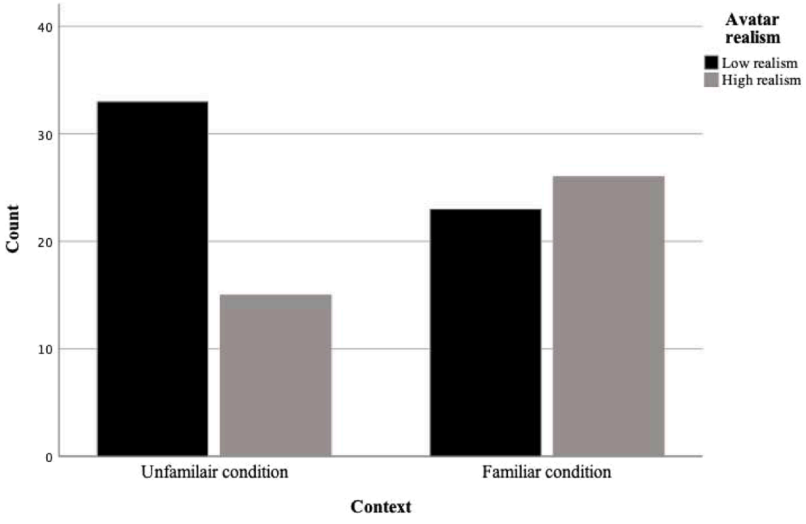


Fig. 3. Users' preference for avatar realism under different familiarity conditions.

Therefore, offering both high- and low-realism options may help accommodate diverse user preferences. For example, platforms such as Roblox already provide users with multiple avatar options with different levels of realism. Additionally, incorporating AI-powered assistants to recommend avatar styles based on users' social environments may further enhance user satisfaction. In sum, our research highlights the importance of designing avatars not only from the platform's perspective, but from that of the users' self-presentation goals and social contexts. Platforms may benefit from focusing on activating the mechanism of wishful identification and immersive experience through the avatar realism design.

5.3. Limitations and future research

This research is not without limitations. First, although we shift the focus from overall visual characteristics—commonly examined in previous studies—to specific aspects of the facial design (i.e., cartoon-like vs. photo-like design), our scope of design features remains limited. According to the definition of form realism [5], additional design elements—such as dynamic movement—should also be considered. The advent of artificial intelligence technology has further expanded the potential of avatars, enabling increasingly complex designs that incorporate distinct personalities, appearances, and behavioral traits. Therefore, future research should explore additional elements of avatar design, including visual, auditory, and even haptic features. Additionally, researchers may also explore the integrated influence of other avatar characteristics (e.g., gender, style, type) on the user's avatar identification in the virtual worlds.

Second, the manipulation in this research relies solely on static visual elements, lacking interactive elements that better reflect real-world virtual experiences. Although we use identical avatars across the first three studies to minimize biases related to personal appearance differences, future research could improve robustness by incorporating avatars based on participants' own faces, as manipulated by Yang et al. [61]. Future studies are encouraged to incorporate more realistic and interactive scenarios to further improve ecological validity.

Third, the expansion of the virtual product market has created significant business opportunities for many product suppliers. This research only focuses on social virtual world contexts, where users expect more possibilities for self-presentation. However, technological advancements have enabled users to embody avatars in augmented and extended reality (AR/XR) environments. In these contexts, when users interact with real-world entities, social norms might influence their preferences for avatar design, encouraging designs that align with users' real-world identities. Thus, understanding how to design avatars in AR/XR scenarios deserve further exploration.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the author(s) used ChatGPT to improve the readability and language of the paper. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the publication.

CRediT authorship contribution statement

Jiahui Huang: Writing – review & editing, Methodology, Investigation. **Minxue Huang:** Writing – review & editing, Funding acquisition, Conceptualization. **Mengmeng Zhan:** Writing – original draft, Formal analysis, Data curation. **Dawei Guan:** Methodology, Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.im.2025.104185.

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